

Effect stdlib first — don't wrap what Effect already gives you

Default failure mode: Claude writes a Path A wrapper over `node:fs` / `Date` / `crypto` / `child_process` (or adds `date-fns` / `zod` / `lodash`) without checking that Effect core or `@effect/platform` already gives the same shape. Run this catalog before writing any *wrapper* or adding any dep.

Pre-check protocol

Before adding ANY external dependency or writing a Path A wrapper:

1. Grep the catalog above for the need. If listed — use Effect's version.
2. Check `@effect/platform` docs page if it's I/O or platform-shaped.
3. Check `effect/<Module>` if it's pure data / algorithmic.
4. Only then consider Path A wrapping or *an external dep*.

Anti-patterns

ts

```
// DON'T - wrapping node:fs by hand
import { readFile } from 'node:fs/promises'

export class FileSystem extends Effect.Service<FileSystem>('FileSystem', {
  sync: () => ({
    readText: Effect.fn('FileSystem.readText')(function* (path: string) {
      return yield* Effect.tryPromise({
        try: () => readFile(path, 'utf8'),
        catch: (cause) => new FsError({ path, cause }),
      })
    }),
  }),
}) {}

// DO - use @effect/platform/FileSystem; it's already the wrapper, with typed errors and
platform abstraction
import { FileSystem } from '@effect/platform'

const program = Effect.gen(function* () {
  const fs = yield* FileSystem.FileSystem
  const text = yield* fs.readFileString(path)
  return text
})
```

ts

```
// DON'T - manual UTC math
const startOfMonth = new Date(Date.UTC(year, month - 1, 1))
const daysInMonth = new Date(Date.UTC(year, month, 0)).getDate()
const isLeap = (y: number) => (y % 4 === 0 && y % 100 !== 0) || y % 400 === 0

// DO - Effect.DateTime + Duration
import { DateTime, Duration } from 'effect'

const start = DateTime.makeZoned({ year, month, day: 1 }, { timeZone: 'UTC' })
const daysInMonth = DateTime.daysInMonth(start)
```

ts

```
// DON'T - JSON.stringify dedup of structural objects
const uniq = Array.from(
  new Map(items.map((x) => [JSON.stringify(x, Object.keys(x).sort()), x])).values(),
)

// DO - Data.struct gives structural equality; HashSet dedups in O(n)
import { Data, HashSet } from 'effect'

const uniq = HashSet.fromIterable(items.map(Data.struct)).pipe(HashSet.toArray)
```

ts

```
// DON'T - adding date-fns / dayjs for date diffs
import { differenceInDays } from 'date-fns'
const days = differenceInDays(end, start)

// DO - Effect.Duration on DateTime diff
import { DateTime, Duration } from 'effect'
const days = DateTime.distance(start, end).pipe(Duration.toDays)
```

When external dep IS warranted

Effect doesn't cover everything. Legitimate external deps:

- **Binary-format SDKs** (image processing: sharp; PDF: pdf-lib; AV: ffmpeg-static)
- **Vendor SDKs** with no Effect equivalent (Stripe, AWS, OpenAI, YNAB, Anthropic) — wrap Path A then
- **Domain-specific algorithms** Effect doesn't ship (ML models, parsers for niche formats, geo libraries)

For these, Path A wrapping is correct. For everything in the catalog above, it's reinventing.